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SURVEY

Interoperability Mechanisms for System-of-Systems: A Systematic Literature Review

THYAGO FERREIRA VIEIRA, RONALDO R. GOLDSCHMIDT, MARIA C. R. CAVALCANTI, AND RICARDO CHOREN

Instituto Militar de Engenharia (IME), Rio de Janeiro 22290-270, Brazil

Corresponding author: Thyago Ferreira Vieira (thyago.ferreira@ime.eb.br)

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ABSTRACT Systems-of-Systems (SoS) engineering faces the intrinsic challenge of orchestrating cooperation among operationally and managerially independent Constituent Systems (CSs). This paper presents a Systematic Literature Review (SLR) on interoperability and integration mechanisms employed in SoS architectures across multiple application domains. Following the guidelines by Kitchenham and Charters and the PRISMA checklist, 58 primary studies were systematically selected and analyzed, addressing three research questions on 1) the predominant mediation mechanisms and enabling technologies, 2) the architectural patterns that support the independent evolution of CSs, and 3) recurrent technical limitations. The selected studies span ten application domains, including defense, energy, transportation, industry, and smart cities, and cover five families of mediation mechanisms, providing a broad and traceable evidence base for researchers and practitioners seeking to understand the current landscape of SoS integration. The results indicate a hegemony of decentralized, agent-based approaches, with co-simulation platforms, semantic mediators, and middleware gateways also present across diverse domains. Taken together, the findings reveal a fragmented landscape: mechanisms are frequently proposed in isolation and evaluated in narrow domain-specific scenarios, with limited cross-mechanism comparison among primary studies and scarce guidance on how to organize mediators in adaptive SoS architectures. These results expose an open gap for lightweight integration mechanisms capable of operating at the boundaries of constituent systems without requiring development effort from their teams, a direction that remains largely unexplored in the literature.

INDEX TERMS Interoperability, system-of-systems, systematic literature review, mediation mechanisms, agent-based systems.

I. INTRODUCTION

The Systems-of-Systems (SoS) paradigm has consolidated itself as a fundamental architectural approach for integrating systems that are operationally and managerially independent, called constituent systems (CSs), with the goal of providing emergent capabilities that cannot be achieved by the parts in isolation [1], [2]. Characterized by their heterogeneity, CSs often belong to different organizational domains and retain their own objectives, resources, and life cycles [3], [4]. This tension between local autonomy and global cooperation underpins the classical SoS taxonomy (directed, acknowledged, collaborative, and virtual), which varies according to the degree of central authority and the adopted governance mechanisms [3]. Seminal works emphasize that the dependence of constituents, combined with continuous evolution, renders SoS engineering intrinsically complex, especially with respect to interoperability and the preservation of systemic integrity over time [1], [4], [5].

At the core of this complexity lies the challenge of integrating systems that have been developed independently. In many critical domains, CSs are legacy, certified, or proprietary systems, for which modifications to source code or internal architecture are infeasible or prohibitively costly [2], [4].

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Intrusive integration strategies, which require tight coupling or deep adaptations inside the CSs, tend to violate operational sovereignty, invalidate existing certifications, and hinder the independent evolution of constituents [2]. Consequently, the literature shows a shift towards non-intrusive integration mechanisms, where concerns such as communication, semantic translation, and behavioral adaptation are encapsulated in external mediators, preserving the autonomy of the original systems [4], [6].

A critical trade-off emerges from the analyzed literature: traditional wrappers, as described by Thiran et al. [7], provide low-latency connectivity but remain passive, lacking the autonomy to handle dynamic SoS behaviors such as fault recovery. Conversely, full-fledged Multi-Agent Systems (MAS) offer high robustness and autonomy but often impose a significant computational overhead and architectural rigidity [8], which are often incompatible with constrained legacy edges. This dichotomy is one instance of a broader spectrum, in which every boundary mediator trades the autonomy it provides against the runtime cost it imposes and the effort required to verify and evolve it. It points to an open gap for **lightweight, boundary-respecting mediation** (mechanisms light enough for constrained legacy edges, yet autonomous, auditable, and evolvable) and, more fundamentally, to the need for a consolidated understanding of where each mediation family sits on this spectrum. Providing that understanding is the aim of this review.

These mediation mechanisms materialize through a range of technical approaches (agent- and MAS-based coordination, wrappers and adapters, gateways and enterprise service buses, semantic mediators, and model-based and co-simulation environments) each encapsulating integration concerns outside the constituents so as to preserve their autonomy [6], [7], [8], [9], [10]. These families are only introduced here; Section IV characterizes each of them in detail.

Despite a growing body of research, the literature still lacks a consolidated view of the effectiveness and limitations of such mechanisms in SoS integration scenarios. Existing studies tend to focus on specific techniques (e.g., only agents or only ontologies) or narrow application domains. There is a shortage of works that systematically synthesize how different classes of mediators (agents, wrappers, gateways, semantic mediators) are employed in SoS, which enabling technologies are most recurrent, how these mediators are organized into architectures that respect the independent evolution of CSs, and in which domains these solutions have been effectively validated. Although many works report challenges related to complexity, interoperability, scalability, and tooling, there is no consolidated picture of the most recurrent technical limitations or of open research gaps in boundary-respecting integration mechanisms and adaptive architectures for SoS. Moreover, although several recent secondary studies review parts of this landscape, none combines coverage of all mediation families, a PRISMA-based selection protocol, and an explicit SoS focus within a single corpus—the conjunction that Section VII examines in detail and that motivates the present review.

Motivated by these gaps, this paper presents a Systematic Literature Review (SLR) focused on SoS integration mechanisms and adaptive architectures in Systems of Systems (SoS). This study synthesizes evidence from 58 primary studies selected through a rigorous search process across multiple digital libraries and snowballing procedures. The main contributions of this work are:

- **A taxonomy of mediation mechanisms:** a classification of the five families of boundary-respecting mediators identified in the corpus (agents and MAS, model-based, semantic, co-simulation, and middleware and gateways) organized by the *layer* and the *moment* at which mediation occurs, and detailing the internal variations and enabling technologies of each family;
- **A mapping of architectural patterns:** an account of how the surveyed mediators are organized into recurring architectural patterns (hierarchical, decentralized, hybrid, and model-based) and of how each pattern trades local constituent autonomy against SoS-wide coordination; these patterns are used as an organizing lens rather than claimed as novel architectures;
- **A diagnosis of technical challenges:** a synthesis of five recurrent challenges that locates the cost of non-intrusive mediation at specific families and layers, most notably the verification of adaptive, agent-based mediators and the long-term maintenance of the semantic agreements mediators depend on, distinguishing them from the generic difficulties common to any large distributed system.

To ensure rigor and reproducibility, this study adopts the PRISMA protocol (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [11] and follows the guidelines by Kitchenham and Charters for systematic reviews in software engineering [12], adapted to the context of Systems-of-Systems engineering.

The remainder of this paper is organized as follows: Section II presents the theoretical background on SoS and interoperability; Section III details the research methodology; Section IV presents the mediation mechanisms identified (RQ1); Section V discusses the architectural frameworks and patterns (RQ2); Section VI addresses the technical challenges and limitations (RQ3); Section VII positions this SLR in relation to recent related reviews; Section VIII discusses threats to validity; and Section IX presents conclusions and directions for future work.

II. BACKGROUND

A. SYSTEM OF SYSTEMS

The Systems-of-Systems (SoS) paradigm has been widely discussed in the systems engineering community as a fundamental approach for describing complex arrangements of independent systems that collaborate to achieve missions they could not accomplish alone. SoS are defined as a set

of systems that exhibit operational independence, managerial independence, geographic distribution, evolutionary development, and emergent behavior [1]. From this perspective, each Constituent System (CS) preserves its own objectives, resources, and life cycle, and retains utility even outside the SoS context.

Building on this conceptual basis, the U.S. Department of Defense (DoD) categorizes SoS into four canonical types according to authority and governance: directed, acknowledged, collaborative, and virtual [3], [5]. In directed SoS there is a strong central authority and a designated mission; in acknowledged SoS, the central organization has recognized responsibility but limited control over constituent funding and evolution; in collaborative SoS, decisions emerge from negotiations among peers; and in virtual SoS, there is no central managing authority, only loosely coupled interactions around shared interests [3]. This taxonomy is widely adopted in the SoS engineering literature and provides a useful lens to analyze trade-offs between control and autonomy [2], [4].

B. INTEROPERABILITY IN SoS

Integrating operationally independent systems imposes severe constraints. In critical domains such as defense, transportation, energy, and healthcare, many CSs are legacy, certified, or proprietary systems, for which internal modifications may be infeasible or could invalidate safety certifications and contractual guaranties [2], [4]. Traditional integration approaches often assume the possibility of refactoring interfaces or injecting code into components. In a SoS context, such *intrusive* strategies are usually inappropriate because they compromise the ability of CSs to evolve according to their own governance, funding, and life-cycle decisions [5].

These constraints motivate the adoption of non-intrusive integration mechanisms. In this approach, responsibilities such as protocol translation, semantic mediation, and adaptation of interaction logic are delegated to external mediators that operate at the boundaries of the systems. The constituent is treated as a black box: only its published interfaces and observable behavior are used, while the internal implementation remains untouched [2].

In safety-critical SoS, this design choice is often framed as a tension between global and local requirements. When the SoS needs to adapt to new missions or threat models, the behavior required at the SoS level may conflict with constraints imposed by individual CSs. Nielsen et al. [4] highlight this as a recurring engineering problem and argue for *cautious adaptation*, where adaptive logic is placed in wrappers or mediating components instead of directly in the constituents [4]. In this context, non-intrusive mediation becomes a key enabler of incremental evolution: new SoS-level capabilities are realized by extending mediators rather than by repeatedly re-certifying and refactoring legacy systems.

III. RESEARCH METHODOLOGY

To obtain insights into the interoperability mechanisms and challenges among constituent systems in a SoS, we conducted an SLR following the methodological guidance by Kitchenham and Charters [12] and the PRISMA checklist [11]. Figure 1 presents the complete search and study selection workflow and the number of retrieved papers at each stage. The diagram follows the PRISMA 2020 flow, adapted to our protocol in two respects: backward and forward snowballing is reported as an additional identification source and merged into the eligibility assessment, and an explicit quality-assessment gate is applied before final inclusion. A detailed explanation of each task will be described later. The search and study selection process was conducted from January 2016 to April 2026 and yielded a total of 58 studies.

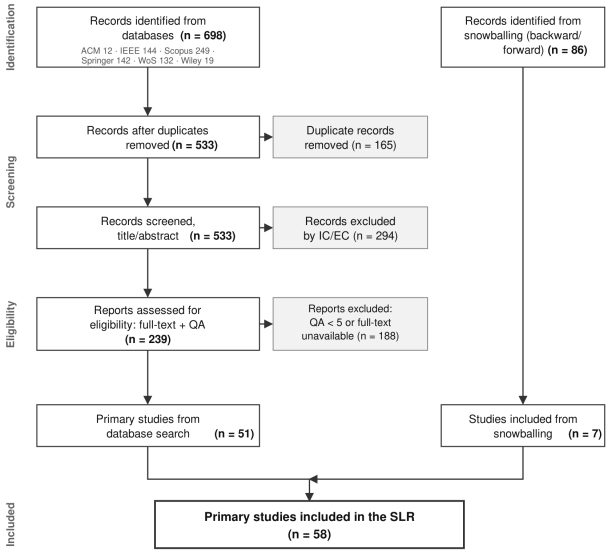


FIGURE 1. Overview of the search, screening, and inclusion process (adapted PRISMA flow).

Because these guidelines were formulated for software engineering in general, we adapted them to the specifics of SoS engineering in four respects. First, the three research questions were framed around SoS-specific constructs (mediation mechanisms, boundary-respecting integration, and the preservation of constituent autonomy), rather than around generic software artefacts. Second, the PICO terms were tailored to the SoS domain: the population combined ''system of systems'' with its common variants (federated, constituent, component, and participating systems), the intervention enumerated boundary-mediation mechanisms (agents, wrappers, middleware, brokers, and gateways), and the outcomes centred on interoperability, non-intrusiveness, and independent evolution; because no keyword set fully captures the heterogeneous vocabulary of the field, the automated search was complemented by backward and forward snowballing. Third, the inclusion and exclusion criteria operationalized the defining properties of SoS (the operational and managerial

independence of constituents [1]), together with the non-intrusiveness constraint, admitting only studies whose mechanisms operate at the boundary of existing systems without internal modification. Fourth, the data-extraction form was instrumented with SoS-specific dimensions, including the governance type of the SoS (directed, acknowledged, collaborative, or virtual) [3], the family of mediation mechanisms, and the architectural pattern, so that extraction characterized how each study addressed interoperability at the SoS level rather than at the level of individual components.

To guide the analysis, we aimed to address the three Research Questions (RQs) presented in Table 1.

TABLE 1. Research questions and motivations.

RQ	Research Question	Motivation
RQ1	What mediation mechanisms and specific technologies are used to enable boundary-respecting integration in SoS?	Identify and classify the main technical approaches used to integrate CSs without requiring internal modifications.
RQ2	What frameworks or architectural patterns are proposed to organize these mediators and support independent evolution?	Understand how boundary mediators are organized at the architectural level and which patterns preserve the autonomy of CSs.
RQ3	What are the reported technical challenges and limitations regarding the use of these mediation mechanisms?	Consolidate recurrent obstacles faced by researchers and practitioners, exposing open research gaps.

A. SEARCH STRATEGY

We began with a search strategy to extract all potentially relevant research papers from academic digital libraries. To design the search string, we utilized the PICO (Population, Intervention, Comparison, Outcomes) framework [12]. The Comparison component was not applicable to our review because our goal was not to conduct a comparison of software with different interventions. Table 2 presents the terms for each PICO component. The search strings were formed through the union of the PICO components and we partially remodeled the search string to match the differences in the search capabilities of each database.

TABLE 2. PICO search terms.

Category	Search Terms
Population	"system of systems", "SoS", "federated system*", "constituent system*", "component system*", "participating system*".
Intervention	"agent*", "multi-agent", "MAS", "adapter*", "wrapper*", "mediat*", "middleware", "broker", "proxy", "gateway".
Outcomes	"integrat*", "interoperab*", "orchestrat*", "coordinat*", "compos*", "non-intrusive", "independent evolution", "decoupling".

To maximize coverage and reduce database bias, six major digital libraries frequently used in software and systems engineering were consulted. The selection was based on their thematic relevance and complementarity: ACM Digital Library and IEEE Xplore were selected as the primary sources for high-impact research in software engineering, computer science, and systems architecture; Scopus and Web of Science were included for their extensive multidisciplinary indexing; and SpringerLink and Wiley Online Library were consulted to encompass major publishers of key journals and book series specialized in systems engineering, cyber-physical systems, and industrial automation.

The automated searches covered the period from January 1, 2016 to April 1, 2026. This ten-year window is a deliberate scoping decision rather than an arbitrary cut-off. It aligns with the review's focus on the *contemporary* generation of non-intrusive mediation mechanisms, which emerged and consolidated after the maturation of the enabling paradigms that reshaped SoS integration: service-oriented and microservice architectures, the Internet of Things, and learning-based agents. Concentrating the primary corpus on this period targets the mechanisms and challenges that are currently active, rather than solution spaces that earlier surveys already document. The window does not, however, exclude foundational work: classical protocol-level mediation (enterprise service buses, wrappers, adapters, and gateways), predates it and is treated throughout this review as consolidated infrastructure, cited as background context [7], [9], [13] rather than re-surveyed. To ensure that seminal pre-2016 contributions informing the corpus were not lost, backward snowballing (described below) explicitly reached older SoS integration proposals, so that the resulting set of 58 primary studies was determined by relevance and quality, not by publication date alone. Table 3 summarizes the number of records imported from each digital library before deduplication.

TABLE 3. Digital libraries consulted and imported records.

Digital Library	Records
ACM Digital Library	12
IEEE Xplore	144
Scopus	249
SpringerLink	142
Web of Science	132
Wiley Online Library	19
Total	**698**

To address the specific syntax and field restrictions of each digital library, the base string was adapted as follows. For ACM Digital Library, the search focused on the Abstract field with a specific publication date range. For IEEE Xplore, targeted searches within the Abstract using field tags and wildcards were performed. For Scopus, the ABS tag was used including all three PICO clusters with filters for document type and stage. For SpringerLink, the advanced search was restricted to title and abstract fields using the core population and intervention clusters. For Web of Science, the TS (Topic)

field covered titles, abstracts, and keywords with the full triple-cluster logic. For Wiley Online Library, the advanced search was restricted to the Abstract field with broader population synonyms. All adaptations and search logs were documented in Parsifal to preserve traceability and ensure full transparency of the search process.

B. STUDY SELECTION

We sought to select any paper on the topic of SoS development that proposed or employed any non-intrusive interoperability model for constituent system integration. The initial set of 698 records was exported from the six digital libraries and imported into Zotero. Automatic duplicate detection based on title, DOI, and metadata was followed by manual verification of borderline cases. Only one canonical version of each study was retained.

1) INCLUSION AND EXCLUSION CRITERIA

Table 4 presents the inclusion and exclusion criteria adopted in this study.

TABLE 4. Inclusion and exclusion criteria.

ID	Criterion
Inclusion Criteria	
IC1	The study must explicitly address Systems-of-Systems or federations of autonomous systems.
IC2	The study must propose, analyze, or evaluate integration mechanisms or architectural patterns.
IC3	The integration approach must be non-intrusive, operating through mediators, wrappers, or agents at the boundaries of constituent systems.
IC4	The study must be written in English and published in a peer-reviewed venue.
Exclusion Criteria	
EC1	Studies focusing solely on conceptual definitions or high-level management must be excluded.
EC2	Works discussing generic middleware or IoT without an SoS context must be excluded.
EC3	Studies in which the proposed integration mechanism is intrinsically intrusive must be excluded.
EC4	Non-peer-reviewed material, short abstracts, or presentation slides must be excluded.

We restricted the selection to works strictly situated within the SoS paradigm that consider the operational and managerial independence of constituent systems [1]. We focused on studies providing concrete technical evidence on connectivity and interoperability solutions, limited to mechanisms that operate at the boundaries of existing systems, preserving legacy implementations and certifications without requiring internal modifications [4], [5].

2) QUALITY ASSESSMENT

We conducted the assessment process using a quality checklist adapted from software engineering SLRs [12], [14], presented in Table 5.

For each criterion, evidence was recorded in the extraction form and an overall quality score was computed per study. These scores were not used to exclude studies, but to support sensitivity analysis, verifying whether the trends identified in the results are driven by a small number of low-quality studies. Studies scoring below 5 out of 8 on the quality checklist were excluded.

TABLE 5. Quality assessment checklist.

ID	Quality Criterion
Q1	The SoS context in which the work is situated must be clearly defined.
Q2	The communication, interoperability, or integration challenges must be clearly stated.
Q3	The research objectives or expected contributions must be explicitly defined.
Q4	The proposed integration mechanism or architecture must be described with sufficient technical detail.
Q5	The study must present an evaluation, experiment, case study, or proof-of-concept.
Q6	The evaluation methodology must be appropriate and described with sufficient detail to judge its rigor.
Q7	The evaluation results must be presented clearly and objectively.
Q8	Limitations, threats to validity, or open issues must be explicitly discussed.

3) SNOWBALLING

Following the identification of the 51 primary studies from the search, we performed a manual search phase using backward and forward snowballing to capture relevant works potentially missed by the digital libraries' indexing [15]. The 51 selected papers served as the starting set (seeds) for this process. Backward snowballing was used to identify seminal works and older SoS integration proposals not captured by the 2016-2026 time window, while forward snowballing, performed with the Litmaps citation-mapping tool, identified recent studies citing these key works. This procedure identified 86 additional candidate papers, of which 7 were included in the final review after rigorous IC/EC and quality assessment, resulting in a final corpus of 58 studies.

C. OVERVIEW OF PRIMARY STUDIES

Figure 2 displays the number of selected papers over the years. We observed that this area of research has received greater attention within the last decade, indicating that this is an area undergoing active research at the time of this study. Hence, this review contributes important and timely value to this emerging area by synthesizing the current knowledge of the underlying SoS development studies.

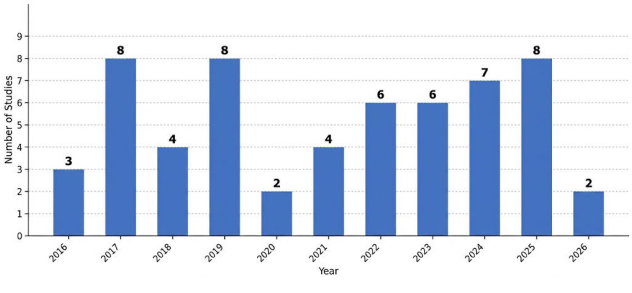


FIGURE 2. Temporal distribution of the 58 primary studies (P1–P58).

D. DATA ANALYSIS

1) DATA EXTRACTION

For each of the 58 primary studies, a structured data extraction form was filled to support traceable answers to RQ1–RQ3 and to enable both quantitative and qualitative synthesis. This form, presented in Table 6, serves as the methodological artifact that guided the collection of bibliographic, technical, and contextual data. All extracted data were stored in dedicated spreadsheets, which subsequently supported the computation of descriptive statistics and the generation of the analytical results presented in Sections IV–VI. This process ensured that distributions of publications, architectural patterns, and validation maturity levels were derived from a consistent and auditable data set.

TABLE 6. Data extraction form (methodological artifact).

Data Category	Extracted Items / Fields
Bibliographic Data	Study ID, Title, Authors, Year, Venue Type, Country/Region.
SoS Context	SoS Type (Directed, Acknowledged, Collaborative, Virtual) [3] and Application Domain.
Mediation (RQ1)	Main Mechanism (Agents/MAS, Wrappers, Gateways, Semantic, Hybrid) and Enabling Technologies (e.g., JADE, HLA, DDS, OPC-UA).
Architecture (RQ2)	Organizational Pattern (Hierarchical, Decentralized, Hybrid) and presence of SoS-specific ADLs or Reference Architectures.
Challenges (RQ3)	Reported Challenges and Limitations (e.g., Scalability, Latency, Determinism, Tooling, Governance).

2) DATA SYNTHESIS

The synthesis combined descriptive statistics with qualitative coding. Descriptive statistics computed counts and percentages per publication year, venue type, country/region, domain, mechanism families, architectural patterns, and validation maturity. Cross-tabulation analyses were performed between mechanism families and application domains (Table 9), between architectural patterns and application domains, and between domains and validation maturity levels, to identify patterns and tensions, notably the dominance of agent-based approaches in defense, the prevalence of co-simulation in energy systems, and the concentration of semantic mediators in environment-related SoS. Qualitative coding captured reported challenges and limitations into thematic categories (e.g., scalability, determinism, interoperability, tooling support). The combination of structured extraction, quality assessment, and mixed synthesis methods provides the basis for the results and interpretations developed in Sections IV–VI.

To keep these categories mutually exclusive, each primary study was assigned to a single mechanism family and a single architectural pattern. When a study combined mechanisms from more than one family, it was classified by its *primary* mediation role, the mechanism carrying the central integration responsibility, following the two explicit criteria, the locus of mediation and the primary mediation role, detailed in Section IV-A. Because every assignment was recorded in the structured data-extraction form, the resulting counts and cross-tabulations are directly traceable to the individual studies behind each category.

IV. MEDIATION MECHANISMS (RQ1)

A. A TAXONOMY OF MEDIATION MECHANISM FAMILIES

Across the 58 primary studies, five main families of mediation mechanisms emerge as central to enabling interoperability among constituent systems in SoS: (i) agent-based mediation, (ii) model-based mediation, (iii) semantic and ontological mediation, (iv) co-simulation and hybrid simulators, and (v) middleware, gateways, and protocol translators. Figure 3 presents this taxonomy, organizing the families according to the layer at which mediation occurs,from communication and protocol translation at the lowest level, through model and simulation federation, up to meaning negotiation and autonomous coordination.

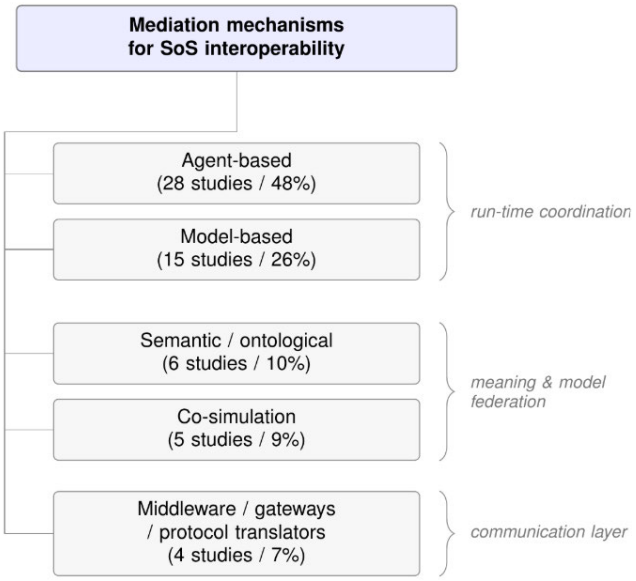


FIGURE 3. Taxonomy of mediation mechanism families identified in the corpus ($n = 58$), with the number and share of primary studies per family, organized by the layer at which mediation occurs.

The classification of each primary study followed two criteria. First, the *locus of mediation*: what is being mediated between constituent systems—messages and protocols (middleware/gateways), engineering models (model-based), simulation executions (co-simulation), meaning and vocabularies (semantic/ontological), or interactions and decisions (agent-based). Second, the *primary mediation role*: when a study combines mechanisms from more than one family (e.g., agents supported by ontologies), it was assigned to the family whose mechanism carries the central mediation responsibility in the proposed architecture. Each study was therefore assigned to exactly one primary family, which preserves the totals reported in Table 8.

TABLE 7. Overview of the 58 Primary Studies: Domain, Mediation Mechanism, Architectural Pattern, and Validation Type.

ID	First Author	Year	Dom.	Mech / Pat / Val	ID	First Author	Year	Dom.	Mech / Pat / Val
P1	Abdoli et al.	2019	Ind	MB / H / Con	P29	Mendes et al.	2018	Gen	Ag / D / Pro
P2	Altarabsheh et al.	2019	Env	Ag / D / Sim	P30	Wanderley et al.	2018	Gen	MB / MB / Con
P3	Arasteh et al.	2016	En	MB / H / Sim	P31	Naeem et al.	2025	Tra	CS / D / Sim
P4	Barbierato et al.	2022	En	CS / H / Sim	P32	Nomaguchi & Fujita	2017	Gen	Ag / H / Sim
P5	Behmer et al.	2016	Def	Ag / Hy / Con	P33	Oquendo	2019	Tra	MB / MB / Con
P6	Binder et al.	2019	En	CS / H / Sim	P34	Oquendo	2023	Tra	MW / Hy / Con
P7	Boldrini et al.	2025	Env	Sem / H / Pro	P35	Palensky et al.	2017	En	CS / H / Sim
P8	Bouskela et al.	2022	Gen	MB / MB / Sim	P36	Patolla et al.	2021	Ind	MW / H / Pro
P9	Bouziat et al.	2017	Gen	Ag / D / Sim	P37	Qu et al.	2017	Def	Ag / D / Sim
P10	Bucaioni et al.	2023	Gen	MB / MB / Con	P38	Raman & Murugesan	2022	Def	Ag / D / Sim
P11	Cabral et al.	2024	Ind	MW / H / Pro	P39	Raman & Murugesan	2022	Def	Ag / D / Sim
P12	Caillot et al.	2024	Tra	Ag / D / Sim	P40	Raz et al.	2017	Def	Ag / Hy / Con
P13	Elhabbash et al.	2020	Gen	Sem / MB / Pro	P41	Regueiro et al.	2017	Env	Sem / MB / Pro
P14	Esmaeilzadeh et al.	2019	Tra	Ag / Hy / Sim	P42	Zhang R. et al.	2019	Def	MB / D / Sim
P15	Fang et al.	2025	Gen	MB / MB / Sim	P43	Sadik et al.	2023	Def	Ag / Hy / Sim
P16	Feng et al.	2023	Gen	MB / H / Sim	P44	Seitz et al.	2021	Ind	Ag / H / Pro
P17	Figueiredo et al.	2025	Sur	MB / MB / Con	P45	Sheikh & Chong	2026	SC	Ag / Hy / Sim
P18	Gomes Jr. et al.	2023	Gen	MW / D / Pro	P46	Shrivastava et al.	2024	Env	Ag / Hy / Sim
P19	Graciano Neto et al.	2017	Gen	MB / MB / Sim	P47	Singh et al.	2022	Rob	Ag / Hy / Sim
P20	Guariniello et al.	2020	Spa	Ag / H / Con	P48	Soyez et al.	2017	Gen	Ag / H / Sim
P21	Guariniello et al.	2021	Spa	MB / H / Con	P49	Umair et al.	2021	En	Ag / H / Sim
P22	Hofmeister et al.	2024	Env	Sem / D / Pro	P50	Villas et al.	2024	Gen	Ag / H / Con
P23	Kawambwa & Mnyang.	2023	En	Ag / H / Sim	P51	Wanderley et al.	2016	Ind	Ag / H / Pro
P24	Kumar et al.	2018	Gen	MB / H / Sim	P52	Wanderley et al.	2018	Ind	Ag / H / Pro
P25	Liu et al.	2024	Tra	Ag / D / Sim	P53	Ye et al.	2025	Def	Ag / D / Sim
P26	Maia et al.	2019	Gen	Ag / Hy / Con	P54	Younes et al.	2019	Def	Sem / MB / Con
P27	Manjah et al.	2024	Sur	Ag / D / Pro	P55	Yuan et al.	2025	Def	Ag / D / Sim
P28	Melén et al.	2025	Ind	CS / H / Sim	P56	Zhang L. et al.	2022	Ind	MB / MB / Pro
P57	Mogahed & Mansouri	2026	Gen	Sem / MB / Pro	P58	Torkjazi et al.	2026	Gen	MB / H / Con

Domains: Def=Defense; En=Energy; Tra=Transportation; Ind=Industry; SC=Smart City; Rob=Robotics; Gen=General SoS; Env=Environment; Spa=Space; Sur=Surveillance.
Mechanisms: Ag=Agent/MAS; MB=Model-based; CS=Co-simulation; Sem=Semantic/Ontological; MW=Middleware/Gateway.
Patterns: H=Hierarchical; D=Decentralized; Hy=Hybrid; MB=Model-based.
Validation: Sim=Simulation; Pro=Prototype/PoC; Con=Conceptual.

TABLE 8. Distribution of primary studies across classification dimensions ($n = 58$).

Domain	n	%	Mech.	n	%
General SoS	17	29	Agent/MAS	28	48
Defense	10	17	Model-based	15	26
Industry	8	14	Co-simulation	5	9
Energy	6	10	Semantic/Onto.	6	10
Transportation	6	10	Middleware/GW	4	7
Environment	5	9			
Other[†]	6	10			
Pattern	n	%	**Validation**	n	%
Hierarchical	22	38	Simulation	30	52
Decentralized	15	26	Conceptual	14	24
Model-based	12	21	Prototype/PoC	14	24
Hybrid	9	16	Field Deployment	0	0

[†] "Other" comprises Space (2), Surveillance (2), Smart City (1), and Robotics (1); these are detailed in Table 9.

TABLE 9. Cross-tabulation: application domain vs. mediation mechanism ($n = 58$). Bold = dominant mechanism per domain.

Domain	Ag	MB	CS	Sem	MW	n
Defense	**8**	1	0	1	0	10
General SoS	6	**8**	0	2	1	17
Industry	**3**	2	1	0	2	8
Energy	2	1	**3**	0	0	6
Transportation	**3**	1	1	0	1	6
Environment	2	0	0	**3**	0	5
Space	**1**	1	0	0	0	2
Surveillance	**1**	1	0	0	0	2
Smart City	**1**	0	0	0	0	1
Robotics	**1**	0	0	0	0	1
Total	**28**	**15**	**5**	**6**	**4**	**58**

Ag=Agent/MAS; MB=Model-based; CS=Co-simulation; Sem=Semantic/Ontological; MW=Middleware/Gateway.

These families differ not only in the layer at which they operate but also in how they preserve the autonomy of constituent systems. Middleware, gateways, and protocol translators intervene at the boundary of the communication stack, encapsulating syntactic heterogeneity [7], [9]; semantic mediators address vocabulary mismatches and meaning conflicts [10], [16]; co-simulation platforms federate independently developed simulation models through standardized interfaces; model-based approaches mediate at design time, generating integration artifacts from unified representations; and agent-based approaches delegate mediation to autonomous components that coordinate at run time [8]. As Table 8 shows, agent-based mediation dominates the contemporary corpus (28 studies, 48%), followed by model-based approaches (15, 26%), while classical communication-layer mediation appears in a comparatively small share (4, 7%)—a distribution whose implications are analyzed in Section IV-B.

B. ANALYSIS OF MECHANISM FAMILIES

This subsection examines each family in turn, characterizing how its mechanisms mediate between constituent systems, which internal variations appear in the corpus, and which SoS

challenges they address. The families are presented from the communication layer upwards, following the organization of Figure 3.

1) MIDDLEWARE, GATEWAYS, AND PROTOCOL TRANSLATORS

This family intervenes at the lowest mediation layer: constituent systems remain untouched while a dedicated component absorbs the syntactic heterogeneity of their communication stacks. The classical solution space is broad and predates the review window, comprising at least three well-established variations: *protocol gateways and translators*, which convert message formats and transport protocols between otherwise incompatible endpoints; *integration buses* such as the Enterprise Service Bus, which centralize routing, transformation, and mediation logic in a shared infrastructure [9]; and *wrappers and adapters*, which encapsulate a single legacy system behind a modernized interface [7], [13]. These variations differ in where the translation burden is placed (at the edge, in shared infrastructure, or around each constituent) and, consequently, in how integration effort scales with the number of heterogeneous systems.

Within the corpus, four studies adopt this family as their primary mediation mechanism, each occupying a distinct point in that space. The multiprotocol middleware translator of [P11] is an edge-translation solution: it integrates IoT devices and platforms that speak different protocols through a translation layer that preserves local autonomy, and reports a working prototype in an industrial setting. The architecture for automatic integration of IO-Link sensors ([P36]) combines middleware with discovery mechanisms, so that sensors can be plugged into a larger industrial SoS without modifying legacy controllers, a gateway variation in which mediation includes capability discovery, not only format conversion. [P18] extends the family beyond device-level protocols to data-level integration: it harvests, indexes, and enriches missing-person case records scattered across distributed web repositories maintained by independent organizations, treating each repository as a black box accessed only through its public interface, with the integration layer absorbing the heterogeneity of formats and vocabularies, while [P34] explores mediating control components that interpose fuzzy decision logic between vehicle subsystems, extending the translator role from data formats to control actions.

Two observations follow. First, this family concentrates the highest proportion of prototype-level validation in the corpus (three of its four studies report working prototypes), consistent with its proximity to deployable infrastructure. Second, and more significantly, it accounts for only 7% of the primary studies, by far the smallest share among the five families. We interpret this not as evidence that protocol-level mediation is unimportant, but as evidence that it has been largely *consolidated*: the classical middleware solution space matured before the review window [7], [9], and contemporary SoS research treats it as available infrastructure, shifting its attention to mediation problems that remain open at higher layers: meaning negotiation, model federation, and autonomous coordination. Individual studies such as [P11] should therefore be read as domain-specific instantiations within a mature solution space, rather than as representative of what protocol translation can achieve in general. A practical consequence for SoS engineers is that protocol-level heterogeneity is today the most tractable of the integration barriers, whereas the corpus indicates that semantic alignment and coordination logic remain the dominant sources of integration effort.

In summary. Protocol-level mediation spans a broad and mature space (gateways, service buses, and wrappers) that this review treats as consolidated infrastructure rather than a closed topic; despite that maturity, it still surfaces in four studies within the contemporary corpus, underscoring its continued relevance.

2) MODEL-BASED MEDIATION

The second-largest family (15 studies, 26%) shifts mediation from run time to *design time*: instead of a component that brokers messages during operation, the mediator is an explicit model or architectural description from which integration artifacts are derived, verified, or generated. Constituent systems are integrated by being represented in a shared modeling formalism, so that heterogeneity is resolved at the level of models rather than at the level of live communication. Within the corpus, this family spans three internal variations that differ markedly in maturity.

The first variation relies on *formal and notational languages* that describe the SoS architecture and its mediation logic explicitly. Architecture description languages specialized for SoS are the clearest instance: [P33] and [P19] use SosADL, a formal language grounded in the π-Calculus, to declare the communication and mediation components between constituents, with the latter feeding the resulting architecture into a DEVS simulator for behavioral evaluation. Other studies adopt general-purpose or purpose-built modeling languages coupled to execution engines: SysML with the Unified Architecture Framework (UAF) to build executable models and evaluate measures of effectiveness ([P58]); a dedicated integrated modeling-and-simulation language executed on a DEVS-based engine to describe system-level architecture and physical behavior as a whole ([P56]); and the Modelica language, combined with the FORM-L requirements language and FMI-based co-simulation, to bind behavioral, architectural, and requirements models for simulation-driven verification ([P8]). In all these cases, the model itself is the mediator: interoperability is achieved by expressing constituents and their interactions in a common, formally analyzable notation.

The second variation couples model-driven design with *executable validation*, using concrete technologies not as the contribution but as evidence that the model achieves integration in practice. Reference [P10] proposes a model-driven

approach for representing reference architectures and automatically checking the conformance of concrete architectures against them, using an IoT ecosystem as an illustrative case. Similarly, [P17] develops an architectural model for an information SoS and implements a mediator to measure the interoperability levels it can attain, validated through communication technologies such as REST and streaming protocols. Here the model remains the core mediation mechanism, while the implementation serves to verify conformance or demonstrate attainable interoperability. A related methodological contribution is provided by [P30], which builds on the Memory-Broker-Agent (MBA) platform to propose a step-by-step guide (GAMBAD) for developing new SoS in practice, mediating not through a novel artifact but through a reusable development method grounded in an existing broker-based architecture.

The third and largest variation is *purely conceptual*: nearly half of the family ([P1], [P3], [P15], [P16], [P21], [P24], [P42]) proposes SoS frameworks and models that mediate through a unified representation without prescribing any implementation technology. These works operate at the level of conceptual modeling and multi-objective decision frameworks, aimed at understanding and analyzing SoS interactions rather than at producing a deployable integration layer.

This internal spread is itself a finding. Unlike the middleware family, whose solution space is technologically consolidated, the model-based family is dominated by design-time and conceptual mediation: of its 15 studies, only about half name a concrete enabling technology, and fewer still carry that model through to an executable mediator. Model-based mediation therefore addresses a different concern from the communication-layer families: it targets the *architectural coherence* of the SoS and the preservation of independent evolution through explicit models, rather than the runtime exchange of data. This complementarity between design-time model mediation and run-time coordination, examined next, is a recurring structure across the corpus.

In summary. Model-based mediation secures architectural coherence at design time, but only about half of the family carries a model through to an executable mediator—the rest remains conceptual.

3) AGENT-BASED MEDIATION

Agent-based mediation is the largest family in the corpus (28 studies, 48%). Here, mediation is delegated to autonomous software agents that wrap or represent constituent systems and coordinate at run time through negotiation, message exchange, or emergent behavior, preserving the local autonomy of each constituent. The family is internally diverse, and a first observation is decisive for interpreting its dominance: only nine of the 28 studies name a concrete enabling technology, while the majority employ agents as a *modeling and simulation paradigm* rather than as a deployed multi-agent platform. We first characterize the technologically explicit variations and then the larger conceptual body.

Among the studies that name concrete technologies, three variations emerge. The first relies on *classical FIPA-oriented MAS platforms*, in which standardized agent frameworks and communication protocols carry the mediation: [P43] uses JADE with the FIPA-ACL agent communication language, [P48] builds on MaDKit (an Agent-Group-Role platform) together with the IRM4MLS multi-level simulation model, and [P51] employs the OMAS open multi-agent framework. The second variation couples agents to *industrial and process standards*, embedding mediation into automation stacks: [P44] combines OPC-UA with a domain protocol for automatic sensor integration, and [P29] integrates JADE agents with BPMN process models. The third variation provides *organizational methods and frameworks* that structure agent roles and coordination: [P27] applies the ASPECS methodology, [P40] defines a Dynamic Architecture Framework over agent-based simulation, and [P52] introduces the Memory-Broker-Agent (MBA) architecture, in which broker agents mediate between requesters and providers through a Contract-Net protocol. A domain-specific instance is [P14], where agents wrap legacy air-traffic management systems (NTML/TFMS) to mediate across an aviation SoS.

The remaining studies use agents without naming an implementation technology, and their insights cluster into three coherent groups. The most prominent—and the clearest signal of where the field is moving—is *learning-based coordination*, where the mediating logic is a reinforcement-learning policy rather than a fixed protocol: [P38] (DDPG), [P39] and [P45] (multi-agent reinforcement learning), [P47] (Soft Actor-Critic), [P55] (MADDPG), and [P53] (reinforcement learning combined with adversarial game theory) evolve the collective behavior of the SoS through agents that adapt at run time. A second group employs *agent-based simulation to explore emergent behavior* in a specific domain—resilient combat architectures ([P5]), cooperative mobility and traffic ([P25]), distributed energy negotiation ([P23], [P49]), environmental and digital-twin monitoring ([P46]), and operational concept validation ([P50])—using agents as the vehicle to study interactions that static methods cannot capture ([P37]). A third group proposes *conceptual agent architectures and decision frameworks* for SoS composition, cautious adaptation, and robustness analysis ([P2], [P9], [P12], [P20], [P26], [P32]), where the contribution is an organizing model of agent interaction rather than a running system.

Two findings follow. First, the prevalence of agent-based mediation reflects the natural fit between agent autonomy and the operational and managerial independence of constituents: an agent can represent a constituent without requiring changes to it. Second, and more revealing, the dominance is largely one of *paradigm rather than deployment*—most studies mobilize agents conceptually or in simulation, and the contemporary research frontier is shifting from classical FIPA platforms toward learning-based agents that evolve SoS behavior at run time. This shift brings adaptability

but, as Section VI discusses, aggravates the challenges of determinism, verification, and certification.

In summary. Agents dominate in numbers (48%) but chiefly as a modelling paradigm; the deployable frontier—learning-based agents—buys adaptability at the price of verifiability.

4) SEMANTIC AND ONTOLOGICAL MEDIATION

Semantic and ontological mediation (6 studies, 10%) addresses a barrier that the previous families largely leave open: even when systems can exchange messages, they may not share the *meaning* of the data exchanged. This family resolves vocabulary mismatches and conceptual conflicts among constituents by mediating at the level of meaning, typically through ontologies and knowledge representation. It is the most technologically consistent family in the corpus: five of its six studies name a concrete enabling technology, and the Semantic Web stack (OWL, RDF, SPARQL) recurs across them.

Two internal variations are visible. The first uses *ontologies as the direct mediation formalism*, where an ontology captures the shared conceptualization and drives integration or composition. Reference [P13] employs OWL to read the hierarchical structure of constituents and generate the semantic descriptions that enable automated SoS composition; [P57] governs an entire interoperable healthcare infrastructure through a computational ontology, combining OWL 2 DL with domain standards (FHIR, HL7, SNOMED) and rule and constraint languages (SWRL, SHACL); and [P54] (the sole study without a named technology) formalizes ontology-weighted belief functions as a theoretical means to mitigate unreliability between constituents. The second variation combines ontologies with *service- and graph-based infrastructures* to mediate access to heterogeneous, distributed data sources: [P41] couples OGC Sensor Observation Services and the O&M model with OWL and SPARQL to translate queries through a standardized semantic layer; [P22] builds on a dynamic Knowledge Graph (The World Avatar) with OWL and SPARQL to interconnect agents through inference; and [P7] provides a Discovery and Access Broker that harmonizes access to large marine metadata collections through pluggable accessors.

A distinctive trait of this family is its concentration in *data-intensive and standards-governed domains* (environmental monitoring, healthcare, and marine data) where the integration problem is inherently about reconciling independently developed conceptual models rather than transport protocols. As Section VI discusses, this strength comes with a specific cost: ontologies must be maintained and evolved as domains change, which becomes the dominant challenge for this family rather than connectivity or coordination.

In summary. Semantic mediation is the most technology-consistent family and the one that tackles meaning rather than transport; its defining cost is keeping shared ontologies alive as constituents evolve.

5) CO-SIMULATION AND HYBRID SIMULATORS

Co-simulation (5 studies, 9%) mediates at the level of *simulation execution*: rather than integrating the constituent systems themselves, it federates their independently developed simulation models, letting heterogeneous simulators run together under coordinated time synchronization and data exchange. This family is the most standards-driven in the corpus: four of its five studies build on established co-simulation standards, chiefly the High-Level Architecture (HLA) and the Functional Mock-up Interface (FMI).

The dominant variation orchestrates *multiple domain simulators through a co-simulation bus or master*. Reference [P4] integrates simulators written in different programming languages using HELICS (built on the IEEE HLA standard), alongside Mosaik and AIOMAS, and compares their scalability limits for multi-energy systems; [P6] applies Mosaik to design and evaluate integrated smart-grid applications; and [P35] couples HLA (IEEE 1516), FMI, and Ptolemy II to build real-time, hardware-in-the-loop interoperability infrastructures. A second variation embeds co-simulation within *agent-based value exploration*: [P28] uses the Repast Symphony agent toolkit together with MATLAB to exchange data continuously across models during early-stage SoS design. The remaining study ([P31]) proposes an extensible agent-based computational structure for simulating dynamic, collaborative fleet arrangements without committing to a named platform.

Co-simulation is concentrated in the energy domain, where constituents are physical subsystems with mature, vendor-specific simulation models that cannot be merged into a single monolithic simulator. Its reliance on standards such as HLA and FMI makes it the family with the clearest interoperability foundation, but also the one whose mediation is confined to the design and analysis phase: it integrates *models of* constituents rather than the operational systems themselves. This positions co-simulation as complementary to the run-time families, a distinction consolidated in the synthesis that follows.

In summary. Co-simulation federates *models of* constituents rather than the systems themselves: a standards-driven, design-time family concentrated in energy and complementary to the run-time families.

C. SYNTHESIS OF MEDIATION MECHANISMS

The family-by-family analysis reveals a corpus organized along two axes: the *layer* at which mediation occurs (from communication, through model and simulation federation, up to meaning and coordination) and the *moment* of mediation, whether at design time or run time. Read along these axes, the distribution of the 58 studies is itself the central finding of RQ1. Mediation in contemporary SoS research has migrated away from the communication layer: classical middleware and protocol translation, though foundational, account for only 7% of the corpus and are treated as

TABLE 10. Synthesis of mediation mechanism families and their internal variations (RQ1). Each row consolidates studies characterized in Section IV-B. (New table added in revision.)

Family	Internal variation	How it mediates	Primary SoS challenge addressed
Communication layer — run time			
Middleware / Gateway (4)	Protocol/gateway & data-level translation	Edge, bus, or wrapper components absorb syntactic and data-format heterogeneity (P11, P18, P34, P36)	Syntactic and protocol heterogeneity
Model & simulation layer — design time			
Model-based (15)	Formal / notational languages	A shared formal model (SosADL, SysML/UAF, Modelica) is the mediator (P8, P19, P33, P56, P58)	Architectural coherence; independent evolution
	Model-driven with executable validation	Model drives conformance checking or interoperability measurement (P10, P17, P30)	Design-time verification of integration
	Conceptual (no named technology)	Unified conceptual model of SoS interaction (P1, P3, P15, P16, P21, P24, P42)	Understanding and analysis of interactions
Co-simulation (5)	Standards-based orchestration	HLA/FMI federate independent simulators (P4, P6, P35)	Model-level integration of physical subsystems
	Within agent-based value exploration	Agent toolkits exchange data across models (P28, P31)	Early-stage design-space exploration
Meaning layer — run time			
Semantic / Ontological (6)	Ontology-as-formalism	Ontology reconciles vocabularies and meaning (P13, P54, P57)	Semantic and conceptual conflicts
	Ontology + service/graph infrastructure	Semantic layer mediates access to distributed data (P7, P22, P41)	Meaning-level access to heterogeneous data
Coordination layer — run time			
Agent-based (28)	FIPA / classical MAS platforms	Standard agent frameworks and protocols coordinate constituents (P43, P48, P51)	Decentralized coordination
	Agents on industrial/process standards	Agents embedded in automation stacks (P29, P44)	Integration with legacy automation
	Organizational methods / frameworks	Methods structuring agent roles and brokering (P27, P40, P52)	Structured multi-agent organization
	Legacy-domain agent wrapping	Agents wrap legacy domain systems (P14)	Non-intrusive legacy integration
	Learning-based coordination (RL/DRL)	Reinforcement-learning policies coordinate at run time (P38, P39, P45, P47, P53, P55)	Dynamic, adaptive coordination
	Agent-based simulation of emergence	Agents used to study emergent behavior (P5, P23, P25, P37, P46, P49, P50)	Analysis of emergent SoS behavior
	Conceptual agent architectures	Organizing models of agent interaction (P2, P9, P12, P20, P26, P32)	Conceptual SoS composition

consolidated infrastructure, whereas the bulk of research concentrates on higher layers: agent-based coordination (48%) and model-based mediation (26%). Within this shift, a finer frontier is visible: the agent family, dominant in numbers, is moving from classical FIPA platforms toward learning-based agents that evolve SoS behavior at run time.

This layered view also clarifies how each family preserves constituent autonomy. Middleware, semantic, and agent-based mediation operate at run time on the operational systems (absorbing protocol, meaning, and coordination heterogeneity, respectively) while model-based and co-simulation approaches mediate at design time, over *models of* the constituents rather than the systems themselves. The two groups are complementary: design-time families secure architectural coherence and independent evolution before deployment, whereas run-time families sustain interoperability during operation. Table 10 consolidates the internal variations identified for each family, the mechanism through which each mediates, and the primary SoS challenge it addresses.

Complementing this mechanism-oriented synthesis, Table 9 maps how the families distribute across application domains. The two views are read together: while Table 10 characterizes *how* each family mediates, the cross-tabulation shows *where* each is preferentially adopted: agent-based mediation dominating defense and mobility, model-based approaches prevailing in general SoS engineering, co-simulation concentrated in energy, and semantic mediation in environmental and data-intensive domains. Together, they substantiate the answer to RQ1: SoS integration is served not by a single mechanism but by a layered repertoire of families, each addressing a distinct facet of heterogeneity and each preserving constituent autonomy in a different way.

V. FRAMEWORKS AND PATTERNS (RQ2)

The analysis of the primary studies shows that boundary mediators are rarely proposed in isolation. In most cases, they are embedded in broader architectural frameworks that seek to reconcile constituent autonomy with SoS-wide coordination. Four main architectural families emerge from the corpus.

The first family comprises **hierarchical and layered architectures**. Several studies adopt an explicitly layered organisation, with local controllers or mediators at the constituent level and one or more coordination layers above.

In power and energy systems, SoS-based expansion planning and hybrid simulation frameworks follow this pattern ([P3], [P4], [P6], [P23], [P35]), where local controllers wrap physical subsystems while global optimisation and supervisory logic are centralised in higher layers. In manufacturing and Industry 4.0, architectures such as the multi-agent automation platform of [P44] and the MBA model for collaborative work in [P52] also combine local agents with a structured hierarchy of services and coordination roles.

The second family consists of **fully decentralised and peer-to-peer architectures**, which emphasise distributed control in which mediators associated with each constituent coordinate directly via peer-to-peer interactions. This is particularly visible in works dealing with UAV swarms, networked combat systems, and intelligent SoS under uncertainty ([P38], [P39], [P42], [P53], [P55]), where agents embedded in or wrapping each constituent negotiate, learn, or adapt behaviours without a single central orchestrator. In smart-city and mobility scenarios ([P25], [P31]), multi-agent simulations explore emergent behaviour in transportation and urban mobility, reinforcing the prevalence of decentralised patterns in highly dynamic environments.

The third family encompasses **hybrid or hierarchical-decentralised architectures**, which explicitly combine hierarchical and decentralised elements. Local mediators make short-term decisions on behalf of constituents, while higher-level entities define long-term goals, policies, or mission constraints. Examples include resilient littoral combat architectures ([P5]), information-fusion SoS ([P40]), and self-adaptive SoS architectures for unmanned vehicle fleets ([P43], [P45]). These works typically rely on a mixture of local autonomy and global supervision, well suited to domains where safety and regulatory constraints coexist with dynamic operational demands.

the mediation *mechanism* (RQ1: how a mediator resolves heterogeneity) is independent of the architectural *pattern* (RQ2: how control and coordination are distributed among constituents). A study may therefore be model-based in mechanism yet hierarchical in control: [P24], [P28], and [P32] employ explicit models or ontologies as their mediation mechanism but organise control hierarchically, and are counted here under the hierarchical pattern. The family below therefore groups only studies whose *architecture itself* is model-driven, in which a shared model or ontology is the organising backbone rather than one mechanism among others.

The fourth family comprises **model-based and ontology-centered frameworks**, which organize mediators around explicit models or ontologies. Ontology-based SoS composition and mediation architectures ([P13], [P41], [P54]) use semantic models as the backbone for integrating heterogeneous data sources and services. Model-based approaches for SoS architecture and simulation ([P10], [P15], [P19], [P30], [P56]) define explicit architectural or behavioral models from which mediators, stimulus generators, or co-simulation setups are derived, enabling generators and simulators to operate over a unified representation while underlying tools remain independent.

Overall, RQ2 shows that mediator-based integration is rarely a purely technical add-on: mediators are systematically embedded in architectural frameworks that constrain how they can evolve and how much autonomy they can exercise. In domains with strong safety or regulatory requirements, hierarchical and hybrid architectures dominate ([P3], [P35], [P44], [P52]), whereas fully decentralised patterns and model-based frameworks are primarily used in simulation-heavy studies exploring emergent behaviour ([P25], [P31], [P38], [P39], [P42], [P53], [P55]). Table 11 summarises the four architectural families identified, their key characteristics, dominant application domains, and the assignment of all 58 primary studies to their pattern.

VI. TECHNICAL CHALLENGES (RQ3)

Across the primary studies, authors explicitly discuss several technical challenges and limitations that arise when mediators are used to integrate autonomous constituents without modifying their internals. These challenges can be grouped into five recurring themes. What sets them apart from the well-known difficulties of any large-scale distributed system is that each arises specifically from the decision to preserve constituent autonomy through external mediators: they are the price of non-intrusiveness, not of scale alone.

The first and most pervasive challenge concerns the **overhead intrinsic to mediation itself**. This is not the generic observation that large distributed systems are hard to scale; it is the more specific finding that interposing an autonomy-preserving mediator between constituents adds a processing, communication, and coordination cost that would simply not exist under intrusive integration. The corpus shows this cost concentrating in the agent-based and co-simulation families. Studies on multi-agent architectures for energy systems, production systems, and UAV swarms ([P23], [P38], [P39], [P42], [P44], [P49], [P55]) report concerns with scalability as the number of agents, constituents, or controlled resources increases. In co-simulation frameworks ([P4], [P6], [P28], [P31], [P35]), the orchestration of multiple simulators and mediators also leads to synchronisation bottlenecks. These results suggest that the cost of mediation (in the form of latency, resource usage, and coordination complexity) must be explicitly considered in SoS design.

The second challenge relates to **determinism, verification, and certification**. Several works highlight the difficulty of reasoning about the global behaviour of SoS when integration is mediated by autonomous agents, learning-based components, or loosely coupled wrappers. In defence, transport, and energy applications ([P5], [P14], [P35], [P40], [P42], [P53], [P55]), authors point out that emergent behaviour, partial observability, and runtime adaptation complicate formal verification and certification. Deep Reinforcement Learning approaches for SoS behaviour evolution ([P38], [P39]) further reinforce this tension: while they

TABLE 11. Architectural patterns identified across the 58 primary studies (RQ2).

Pattern	*n*	Key Characteristics	Dominant Domains	Primary Studies
Hierarchical	22	Layered control; local mediators at constituent level with centralised coordination above	Energy, Industry	P1, P3, P4, P6, P7, P11, P16, P20, P21, P23, P24, P28, P32, P35, P36, P44, P48, P49, P50, P51, P52, P58
Decentralized	15	Peer-to-peer mediators; no central orchestrator; emergent coordination through local negotiation	Defense, Smart Cities	P2, P9, P12, P18, P22, P25, P27, P29, P31, P37, P38, P39, P42, P53, P55
Hybrid	9	Combined local autonomy and global supervision; policy constraints at higher levels	Defense, Transportation	P5, P14, P26, P34, P40, P43, P45, P46, P47
Model-based	12	Ontology or explicit model as integration backbone; mediators derived from shared representations	General SoS	P8, P10, P13, P15, P17, P19, P30, P33, P41, P54, P56, P57

increase adaptability, they also reduce the predictability of the integrated system as a whole.

The third challenge involves **semantic heterogeneity and ontology management**. In data-intensive SoS, the main integration barrier is not connectivity but semantic alignment. Works on semantic mediation and ontology-based integration ([P7], [P13], [P22], [P41], [P54]) report difficulties in maintaining shared vocabularies over time, evolving ontologies as domains change, and resolving conflicts between independently developed conceptual models. Even when mediators successfully map between data formats, ensuring that all parties share a consistent understanding of concepts and units remains a non-trivial task.

The fourth challenge concerns **tooling, integration effort, and legacy constraints**. Multiple studies emphasise the engineering effort required to design, deploy, and maintain mediators around legacy or proprietary systems. Industrial and IoT-focused works ([P11], [P18], [P36], [P44], [P52], [P56]) note that wrappers and gateways often need to be customised on a case-by-case basis, especially when vendor-specific protocols, safety certifications, or access restrictions are involved. This limits reuse and increases long-term maintenance costs, even when boundary separation is preserved from a code perspective.

The fifth challenge addresses **governance, responsibility, and evolution**. Several modelling and method-oriented papers ([P16], [P19], [P24], [P28], [P30], [P32]) highlight challenges that are partly technical and partly organisational. When mediators encapsulate policies and adaptation logic on the SoS boundary, questions arise about who owns and maintains these components, how changes in one organisation's policies affect others, and how to coordinate evolution across multiple stakeholders. These governance issues are particularly acute in collaborative SoS scenarios ([P17], [P19], [P27], [P29], [P52], [P57]), where no single authority can unilaterally impose integration strategies.

Taken together, the answers to RQ3 indicate that mediator-based integration is not a cost-free solution: mediators help preserve the independence of constituents, but they shift complexity to the integration layer. Crucially, the corpus locates this cost precisely. It falls not on protocol-level connectivity (the middleware family that classical work already consolidated and that accounts for only 7% of the corpus) but on the two higher mediation layers where the open problems now concentrate: the verification and certification of adaptive, agent-based mediators, and the long-term maintenance of the semantic agreements on which mediators depend. This gives a concrete meaning to the *lightweight boundary mediation* that motivates this review: a mediator is lightweight not merely when its runtime footprint is small, but when the autonomy-preserving boundary it establishes remains *auditable* and *evolvable*—when its adaptive behaviour can still be verified, and when the shared ontologies and contracts it relies on can be maintained as constituents evolve independently. The practical implication for SoS engineers is therefore specific: rather than devising new mediation mechanisms, the priority is to pair the run-time autonomy of boundary mediators with design-time verifiability and with explicit governance of their semantic and contractual dependencies. Table 12 consolidates the five challenge categories, their primary root causes, and all the primary studies that report each of them. Of the 58 primary studies, 35 explicitly report at least one of these challenges; the remaining 23 (predominantly proposal or conceptual papers) do not discuss limitations in these terms.

VII. RELATED REVIEWS

The growing complexity of Systems-of-Systems, driven by increasingly heterogeneous operational scenarios, has generated a substantial body of secondary and tertiary reviews in recent years. However, these works tend to address the topic in a partial manner, concentrating on specific subsets of mechanisms, on isolated application domains, or adopting methodological perspectives distinct from SoS engineering. This section positions the present SLR against nine recent secondary and tertiary reviews. For each, we report not only its publication but the span of primary literature it covers, since a review's currency is better gauged by the window of studies it synthesizes than by its publication date alone.

Axelsson [17] conducted a systematic literature review on mediators in SoS and software ecosystems, proposing a unified conceptualization of the role of mediators in constituent collaboration. Covering primary literature from 2007 to 2022, it is the work closest to ours in subject matter and, like ours,

TABLE 12. Recurrent technical challenges identified across the primary studies (RQ3).

Challenge	Root Cause	*n*	Primary Studies
Mediation overhead	Autonomy-preserving mediators add processing and coordination cost absent under intrusive integration	12	P4, P6, P23, P28, P31, P35, P38, P39, P42, P44, P49, P55
Determinism & certification	Emergent and adaptive behaviour reduces global predictability	9	P5, P14, P35, P38, P39, P40, P42, P53, P55
Semantic heterogeneity	Vocabulary and concept mismatches across independently developed CSs	5	P7, P13, P22, P41, P54
Tooling & legacy constraints	Case-by-case customisation burden for legacy and proprietary systems	6	P11, P18, P36, P44, P52, P56
Governance & evolution	Unclear ownership and multi-stakeholder authority gaps over mediator policies	11	P16, P17, P19, P24, P27, P28, P29, P30, P32, P52, P57

spans several mediation families; however, as a workshop-scope study its treatment of enabling technologies and of recurrent technical challenges remains partial, and it does not follow a PRISMA-based selection protocol.

Ashfaq et al. [18] conducted an SLR on runtime composition in dynamic SoS, selecting 80 primary studies from the period 2019–2024; although its corpus does span the main mediation families, the review is organized around the problem of runtime composition rather than around a taxonomy of mediation mechanisms, and it does not adopt a PRISMA protocol. It is thus the comparator closest to our work in coverage, differing chiefly in organizing axis and in methodological formalization.

Maciel et al. [19] produced a tertiary study on interoperability types by synthesizing 37 secondary reviews published between 2012 and 2023; given its tertiary nature and its domain-agnostic scope, SoS appears as background rather than as the organizing lens, and mediation mechanisms are not treated as a systematic taxonomy.

Nilsson et al. [20] presented a survey on AI concepts for dynamic SoS interoperability, covering literature from 1999 to 2021. Its mediation analysis is confined to a single paradigm (artificial intelligence and machine learning) so that families such as middleware, co-simulation, and model-based mediation fall outside its scope.

Cavalcante et al. [21] presented a retrospective of software engineering for SoS with scientometric analysis of 57 papers from the SESoS workshop series (2013–2023); its broad software-engineering-for-SoS scope does surface open research challenges, but it neither details the enabling technologies of specific mediation mechanisms nor organizes them into families.

Adesanya et al. [22] conducted an SLR on systems of twinned systems in the SoS context, covering 2017 to 2024. Although its corpus in fact spans all five mediation families, the review is organized around the convergence of SoS with digital twins, so that the families surface as enabling technologies for twinning rather than as the review's structuring taxonomy. With Ashfaq, it is one of the two most complete comparators, differing from ours in organizing lens and in the absence of a PRISMA protocol.

Piperigkos et al. [23] reviewed distributed intelligence in industrial and automotive cyber-physical systems over the 2013–2023 window. It frames its subject as cyber-physical systems-of-systems (CPSoS) but does not adopt the SoS paradigm in the Maier/DoD sense as its organizing lens, and its coverage centres on distributed-intelligence layers rather than on a taxonomy of mediation families.

Maldonado et al. [24] present a survey on multi-agent systems published in IEEE Access, covering 2016 to 2022. It offers broad coverage of multi-agent systems but outside the SoS context: the operational and managerial independence of constituents is not its concern, it addresses a single paradigm rather than multiple mediation families, and its treatment of integration challenges is only incidental.

Finally, Abdelmoumen et al. [25] investigated resilience and adaptability in IoT-based SoS, combining bibliometric and narrative analysis across a broad 2010–2025 window. The review explicitly follows a semi-systematic rather than a PRISMA protocol, and it is organized around resilience and dynamic reconfiguration, so that mediation families appear only as isolated implementation examples rather than as a systematized taxonomy.

Table 13 summarizes the comparative analysis across five dimensions, while Table 14 lists the concrete enabling technologies reported by each review.

The comparative analysis reveals that none of the related works satisfies all the evaluated dimensions simultaneously. Several recent reviews (most notably Ashfaq and Adesanya) do span multiple mediation families and diagnose technical challenges; what distinguishes the present work is therefore not any single dimension but their conjunction. Ours is the only review organized as a PRISMA-based taxonomy of mediation mechanism families (agents and MAS, middleware and gateways, co-simulation, semantic mediation, and model-based approaches) spanning the full interoperability space in the explicit context of SoS, rather than being scoped to runtime composition, digital twins, artificial

TABLE 13. Comparison of the present review with nine related secondary and tertiary reviews, across coverage period and five methodological dimensions. ✓ = addressed; ◐ = partially addressed; × = not addressed.

Review	Period	SoS context	Multiple families	Mechanism/tech. detail	Challenge diagnosis	PRISMA protocol
R1	2007–2022	✓	✓	◐	◐	×
R2	2019–2024	✓	✓	✓	✓	×
R3	2012–2023	◐	◐	✓	✓	×
R4	1999–2021	✓	◐	✓	✓	×
R5	2013–2023	✓	◐	×	✓	×
R6	2017–2024	✓	✓	✓	✓	×
R7	2013–2023	◐	◐	✓	✓	×
R8	2016–2022	×	×	×	◐	×
R9	2010–2025	✓	◐	✓	✓	×
Ours	**2016-2026**	✓	✓	✓	✓	✓

R1: Axelsson [17]; R2: Ashfaq et al. [18]; R3: Maciel et al. [19]; R4: Nilsson et al. [20]; R5: Cavalcante et al. [21]; R6: Adesanya et al. [22]; R7: Piperigkos et al. [23]; R8: Maldonado et al. [24]; R9: Abdelmoumen et al. [25]. *Period* denotes the span of primary literature covered by each review.

TABLE 14. Enabling technologies reported by each related review. The identifiers R1–R9 follow the mapping in Table 13; "Not Applied" denotes reviews that do not name concrete enabling technologies.

Review	Enabling technologies
R1	SosADL
R2	Protégé, RDF/OWL, OPC-UA, MQTT, SysML, SosADL, HLA/RTI, JADE
R3	OWL, RDF, SPARQL, SSN
R4	RDF, FIPA-ACL
R5	Not Applied
R6	OPC-UA, IEC 63278 (AAS), SysML, OWL, DEVS, FMI/FMU
R7	SSD, YOLO, EfficientDet, Faster R-CNN
R8	Not Applied
R9	SHSA, LTL/CTL, Kalman filters

intelligence, or resilience. It also synthesizes the most recent window among the comparators (2016–2026). The systematic diagnosis of five recurrent categories of technical challenges, derived directly from qualitative synthesis of the primary studies, additionally provides a structured agenda for future research on the integration of independent constituent systems.

VIII. THREATS TO VALIDITY

As with any Systematic Literature Review, the findings of this study are subject to validity threats that may affect their completeness or generalizability. This section discusses the main issues following common classifications in secondary studies on software and systems engineering [11], [12].

Construct validity concerns whether the review captured the intended phenomenon: non-intrusive integration mechanisms in Systems-of-Systems. The main threat lies in the formulation of the search string and in the operational definition of "non-intrusiveness" and "SoS", as terminology in the area is heterogeneous: some authors explicitly use the term SoS, while others describe similar arrangements as cyber-physical ecosystems, federations, or networks of systems. To mitigate this threat, the search strategy was derived from a PICO formulation and applied across multiple digital libraries, complemented by backward and forward snowballing. Nonetheless, it is possible that relevant studies employing different terminology were not retrieved.

Internal validity relates to potential biases in study selection and data extraction. Decisions about whether a paper satisfies the SoS and non-intrusiveness criteria can involve subjective judgment, especially in borderline cases such as large distributed systems that do not explicitly claim an SoS perspective. To reduce this risk, explicit inclusion and exclusion criteria were defined a priori and applied consistently throughout the screening phases, and data extraction relied on a structured form aligned with the research questions. Even so, some degree of subjectivity in classification (for example, when assigning the main mediation mechanism or architectural pattern) cannot be completely eliminated.

External validity concerns the extent to which the results can be generalized beyond the corpus analyzed. The primary studies are concentrated in certain domains, notably defense, energy, transportation, industry, and smart cities, and in a subset of countries, which means the synthesis may not fully represent domains such as healthcare or finance, where different constraints and standards apply. Moreover, the automated search was bounded to the period from 2016 to 2026, and only peer-reviewed publications in English were considered. These decisions are common in SLRs but limit the inclusion of very recent preprints, non-English work, and industrial reports.

Conclusion validity relates to the robustness of the inferences drawn from the extracted data. Patterns such as the prevalence of agent-based mediators or the dominance of simulation-based validation are supported by explicit counts, whereas other observations (such as the architectural tension between decentralized and hierarchical patterns) are derived qualitatively. To mitigate over-interpretation, descriptive statistics were always reported together with qualitative synthesis, and strong causal claims were avoided. As in any interpretive study, alternative readings of the same evidence are possible, and the conclusions should be understood as informed by, but not mechanically determined by, the data.

IX. CONCLUSION

Interoperability in Systems-of-Systems engineering is a central challenge, where constituent systems must collaborate

to provide emergent capabilities without compromising their operational and managerial independence.

We have carried out a Systematic Literature Review of 58 primary studies by following well-established SLR guidelines [12] and the PRISMA reporting checklist [11]. The main aim of this study was to identify and synthesize the mediation mechanisms, architectural patterns, and technical challenges reported in the SoS integration literature, addressing three research questions on mechanisms and technologies (RQ1), frameworks and architectural patterns (RQ2), and recurrent technical challenges and limitations (RQ3).

We have found that agent-based mediation is the dominant approach for boundary-respecting integration in SoS, followed by middleware and gateway solutions, co-simulation platforms, and semantic mediators. Architecturally, hierarchical and hybrid patterns prevail in safety-critical domains such as energy and manufacturing, while decentralized and model-based frameworks are more common in simulation-oriented studies. Across all domains, five recurring technical challenges emerge: scalability and performance overhead, determinism and certification, semantic heterogeneity, tooling and legacy constraints, and governance and evolution. As shown in Table 8, none of the 58 primary studies report field deployment as their primary validation method, with 52% relying on simulation and 24% on prototype evaluation, a persistent gap that further limits the transferability of current proposals to operational environments.

Table 13 positions this review in comparison with nine recent secondary and tertiary reviews, covering their respective time periods and five methodological dimensions. Several comparative reviews cover multiple mediation families or diagnose technical challenges, but none combines all five dimensions in a single corpus during the period of this review (2016-2026). The results of this review provide researchers and practitioners with a structured overview of the current state of research on Systems of Systems (SoS) integration, consolidating evidence into a coherent taxonomy of mechanisms, patterns, and challenges. By exposing open research gaps (particularly the need for lighter, standardized, and resilient mediator designs), this Systematic Literature Review (SLR) offers a concrete agenda for future work on Systems of Systems integration.

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The generative AI models Claude and NotebookLM were used solely as supporting tools, in assisting with the translation and language revision of the manuscript text, and supporting the extraction and classification of data from the primary studies. All AI-assisted output was reviewed and verified by the authors.

The complete data extraction spreadsheet supporting the results of this review is publicly available at: https://github.com/comp-ime-eb-br/S2C2-IME/tree/main/deliverables/SLR_spreadsheets

They declare no conflict of interest.

The authorship and full responsibility for the final content, including plagiarism verification and corrections, remain with them.

REFERENCES

[1] M. W. Maier, "Architecting principles for systems-of-systems," *Syst. Eng.*, vol. 1, no. 4, pp. 267–284, 1998.
[2] M. Jamshidi, Ed., *Systems of Systems Engineering: Principles and Applications*. Boca Raton, FL, USA: CRC Press, 2008.
[3] J. S. Dahmann and K. J. Baldwin, "Understanding the current state of U.S. defense systems of systems and the implications for systems engineering," in *Proc. 2nd Annu. IEEE Syst. Conf.*, Apr. 2008, pp. 1–7.
[4] C. B. Nielsen, P. G. Larsen, J. Fitzgerald, J. Woodcock, and J. Peleska, "Systems of systems engineering: Basic concepts, model-based techniques, and research directions," *ACM Comput. Surv.*, vol. 48, no. 2, pp. 18:1–18:41, 2015.
[5] Undersecretary of Defense [AT&L]. (2008). *Systems Engineering Guide for Systems of Systems, Version 1.0*. Office Deputy Under Secretary of Defense for Acquisition and Technology, Systems and Software Engineering, Washington, DC, USA. [Online]. Available: https://www.cto.mil/wp-content/uploads/2024/06/DoD-SE-for-SoS-2008.pdf
[6] F. Oquendo, "Formally describing the software architecture of systems-of-systems with SosADL," in *Proc. 11th Syst. Syst. Eng. Conf. (SoSE)*, Jun. 2016, pp. 1–6.
[7] P. Thiran, J.-L. Hainaut, G.-J. Houben, and D. Benslimane, "Wrapper-based evolution of legacy information systems," *ACM Trans. Softw. Eng. Methodol.*, vol. 15, no. 4, pp. 329–359, Oct. 2006.
[8] A. Dorri, S. S. Kanhere, and R. Jurdak, "Multi-agent systems: A survey," *IEEE Access*, vol. 6, pp. 28573–28593, 2018.
[9] D. A. Chappell, *Enterprise Service Bus*. Sebastopol, CA, USA: O'Reilly Media, 2004.
[10] L. Garcés, F. Oquendo, and E. Y. Nakagawa, "Software mediators as first-class entities of systems-of-systems software architectures," *J. Brazilian Comput. Soc.*, vol. 25, no. 1, p. 8, Dec. 2019.
[11] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, and D. Moher, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, Mar. 2021.
[12] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Keele Univ. Durham Univ. Joint Rep., Keele, U.K., Tech. Rep. EBSE-2007-01, 2007. [Online]. Available: https://www.cs.keele.ac.uk/pages/academics/kitchenham/EBSE2007.pdf
[13] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design Patterns: Elements Reusable Object-Oriented SofSoftware*. Reading, MA, USA: Addison-Wesley, 1994.
[14] K. Petersen, S. Vakkalanka, and L. Kuzniarz, "Guidelines for conducting systematic mapping studies in software engineering: An update," *Inf. Softw. Technol.*, vol. 64, pp. 1–18, Aug. 2015.
[15] C. Wohlin, "Guidelines for snowballing in systematic literature studies and a replication in software engineering," in *Proc. 18th Int. Conf. Eval. Assessment Softw. Eng.*, May 2014, pp. 1–10.
[16] J. Euzenat and P. Shvaiko, *Ontology Matching*. Berlin, Germany: Springer, 2007.
[17] J. Axelsson, "Mediators in systems-of-systems and ecosystems: A systematic literature review and conceptualization," in *Proc. 12th ACM/IEEE Int. Workshop Softw. Eng. Syst. Syst. Softw. Ecosyst.*, Apr. 2024, pp. 21–28.
[18] M. Ashfaq, A. R. Sadik, T. Das, M. Waseem, N. Mäkitalo, and T. Mikkonen, "Runtime composition in dynamic system of systems: A systematic review of challenges, solutions, tools, and evaluation methods," *J. Syst. Softw.*, vol. 232, Feb. 2026, Art. no. 112661.
[19] R. S. P. Maciel, P. H. D. Valle, K. S. Santos, and E. Y. Nakagawa, "Systems interoperability types: A tertiary study," *ACM Comput. Surv.*, vol. 56, no. 10, pp. 1–37, Oct. 2024.
[20] J. Nilsson, S. Javed, K. Albertsson, J. Delsing, M. Liwicki, and F. Sandin, "AI concepts for system of systems dynamic interoperability," *Sensors*, vol. 24, no. 9, p. 2921, May 2024.
[21] E. Cavalcante, T. Batista, and F. Oquendo, "Looking back and forward: A retrospective and future directions on software engineering for systems-of-systems," *J. Softw., Evol. Process*, vol. 36, no. 10, Oct. 2024, Art. no. e2697.
[22] F. Adesanya, K. C. Silva, V. V. G. Neto, and I. David, "Systems of twinned systems: A systematic literature review," 2025, *arXiv:2505.19916*.

[23] N. Piperigkos, A. Gkillas, G. Arvanitis, S. Nousias, A. Lalos, A. Fournaris, P. Radoglou-Grammatikis, P. Sarigiannidis, and K. Moustakas, "Distributed intelligence in industrial and automotive cyber–physical systems: A review," *Frontiers Robot. AI*, vol. 11, Oct. 2024, Art. no. 1430740.

[24] D. Maldonado, E. Cruz, J. Abad Torres, P. J. Cruz, and S. D. P. G. Benitez, "Multi-agent systems: A survey about its components, framework and workflow," *IEEE Access*, vol. 12, pp. 80950–80975, 2024.

[25] A. Abdelmoumen, Z. Benzadri, and I. B. Rodriguez, "Comprehensive exploration of resilience and adaptability in IoT-based system-of-systems: Bibliometric and full-text literature analysis," in *Proc. Softw. Archit. ECSA Tracks Workshops*, 2025, pp. 243–258.

PRIMARY STUDIES

[P1] S. Abdoli, S. Kara, and M. Hauschild, "An investigation into environmental impact of system of systems," *Proc. CIRP*, vol. 80, pp. 584–589, Jan. 2019.

[P2] A. Altarabsheh, A. Kandil, D. Abraham, D. DeLaurentis, and M. Ventresca, "System of systems approach for maintaining wastewater system," *J. Comput. Civil Eng.*, vol. 33, no. 3, May 2019, Art. no. 04019022.

[P3] H. Arasteh, M. S. Sepasian, V. Vahidinasab, and P. Siano, "SoS-based multiobjective distribution system expansion planning," *Electr. Power Syst. Res.*, vol. 141, pp. 392–406, Dec. 2016.

[P4] L. Barbierato, P. Rando Mazzarino, M. Montarolo, A. Macii, E. Patti, and L. Bottaccioli, "A comparison study of co-simulation frameworks for multi-energy systems: The scalability problem," *Energy Informat.*, vol. 5, no. S4, p. 53, Dec. 2022.

[P5] J. Behmer, K. Ogunsina, P. Shah, and S. Srinivasan, "An agent-based modeling approach to creating more resilient littoral combat architectures," in *Proc. IEEE Aerosp. Conf.*, Mar. 2016, pp. 1–9.

[P6] C. Binder, M. Fischinger, L. Altenhuber, D. Draxler, G. Lastro, and C. Neureiter, "Enabling architecture based co-simulation of complex smart grid applications," *Energy Informat.*, vol. 2, no. S1, p. 10, Sep. 2019.

[P7] E. Boldrini, R. Roncella, F. Papeschi, P. Mazzetti, D. Schaap, P. Thijsse, P. Weerheim, and S. Nativi, "Blue-cloud DAB: Developing a platform to harmonize, assess and disseminate marine metadata collections," *Int. J. Data Sci. Anal.*, vol. 20, no. 4, pp. 3317–3336, Oct. 2025.

[P8] D. Bouskela, A. Falcone, A. Garro, A. Jardin, M. Otter, N. Thuy, and A. Tundis, "Formal requirements modeling for cyber-physical systems engineering: An integrated solution based on FORM-L and modelica," *Requirements Eng.*, vol. 27, no. 1, pp. 1–30, Mar. 2022.

[P9] T. Bouziat, S. Combettes, V. Camps, and J. Boes, "Cooperative multi-agent approach for computational systems of systems architecting," in *Proc. 9th Int. Conf. Agents Artif. Intell.*, Porto, Portugal, Feb. 2017, pp. 174–181.

[P10] A. Bucaioni, A. Di Salle, L. Iovino, I. Malavolta, and P. Pelliccione, "Reference architectures modelling and compliance checking," *Softw. Syst. Model.*, vol. 22, no. 3, pp. 891–917, Jun. 2023.

[P11] B. Cabral, R. Venâncio, P. Costa, T. Fonseca, L. L. Ferreira, R. Severino, and A. Barros, "Multiprotocol middleware translator for IoT," in *Proc. 27th Euromicro Conf. Digit. Syst. Design (DSD)*, Aug. 2024, pp. 327–334.

[P12] A. Caillot, S. Ouerghi, Y. Dupuis, P. Vasseur, and R. Boutteau, "Multi-agent cooperative camera-based semantic grid generation," *J. Intell. Robotic Syst.*, vol. 110, no. 2, p. 64, Apr. 2024.

[P13] A. Elhabbash, V. Nundloll, Y. Elkhatib, G. S. Blair, and V. S. Marco, "An ontological architecture for principled and automated system of systems composition," in *Proc. IEEE/ACM 15th Int. Symp. Softw. Eng. Adapt. Self-Manag. Syst. (SEAMS)*, May 2020, pp. 85–95.

[P14] E. Esmaeilzadeh, M. W. Grenn, and B. Roberts, "An agent-based model for improved system of systems decision making in air transportation," *Syst. Eng.*, vol. 22, no. 1, pp. 20–42, Jan. 2019.

[P15] Z. Fang, D. Chen, Q. Ju, and J. Wang, "Architecting path selection method for incremental evolution in system-of-systems," *IEEE Syst. J.*, vol. 19, no. 2, pp. 636–647, Jun. 2025.

[P16] Y. Feng, C. Zhou, Q. Zou, Y. Liu, J. Lyu, and X. Wu, "A goal-based approach for modeling and simulation of different types of system-of-systems," *J. Syst. Eng. Electron.*, vol. 34, no. 3, pp. 627–640, Jun. 2023.

[P17] R. D. C. Figueiredo, B. Mane, D. R. Clavijo, S. R. D. Silva, F. D. A. Moraes, C. Nery, A. P. Magalhães, R. S. Maciel, D. B. Claro, M. V. Dos Santos e Souza, G. M. de Araújo Lima, and J. P. Leite, "Towards a SoIS model for university surveillance," in *Proc. IEEE/ACM 13th Int. Workshop Softw. Eng. Syst. Syst. Softw. Ecosyst. (SESoS)*, May 2025, pp. 25–32.

[P18] J. Gomes, H. S. Bernardino, J. F. de Souza, and E. Rajabi, "Indexing, enriching, and understanding Brazilian missing person cases from data of distributed repositories on the web," *AI Soc.*, vol. 38, no. 2, pp. 565–579, Apr. 2023.

[P19] V. V. G. Neto, C. E. B. Paes, L. Garcés, M. Guessi, W. Manzano, F. Oquendo, and E. Y. Nakagawa, "Stimuli-SoS: A model-based approach to derive stimuli generators for simulations of systems-of-systems software architectures," *J. Brazilian Comput. Soc.*, vol. 23, no. 1, p. 13, Dec. 2017.

[P20] C. Guariniello, T. B. Marsh, R. Porter, C. Crumbly, and D. A. DeLaurentis, "Artificial intelligence agents to support data mining for SoS modeling of space systems design," in *Proc. IEEE Aerosp. Conf.*, Mar. 2020, pp. 1–11.

[P21] C. Guariniello, T. B. Marsh, T. Diggelmann, and D. A. DeLaurentis, "System-of-systems methods for technology assessment and prioritization for space architectures," in *Proc. IEEE Aerosp. Conf.*, Mar. 2021, pp. 1–13.

[P22] M. Hofmeister, J. Bai, G. Brownbridge, S. Mosbach, K. F. Lee, F. Farazi, M. Hillman, M. Agarwal, S. Ganguly, J. Akroyd, and M. Kraft, "Semantic agent framework for automated flood assessment using dynamic knowledge graphs," *Data-Centric Eng.*, vol. 5, p. e14, Jan. 2024.

[P23] S. Kawambwa and D. Mnyanghwalo, "A multi-agent-based symbiotic organism search algorithm for DG coordination in electrical distribution networks," *J. Electr. Syst. Inf. Technol.*, vol. 10, no. 1, p. 6, Jan. 2023.

[P24] P. Kumar, R. Merzouki, and B. O. Bouamama, "Multilevel modeling of system of systems," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 48, no. 8, pp. 1309–1320, Aug. 2018.

[P25] J. Liu, J. Liu, and M. Zhang, "Effective evolutionary principles for system-of-systems: Insights from agent-based modeling in vehicular networks," *Systems*, vol. 12, no. 3, p. 98, Mar. 2024.

[P26] P. H. Maia, L. Vieira, M. Chagas, Y. Yu, A. Zisman, and B. Nuseibeh, "Cautious adaptation of defiant components," in *Proc. 34th IEEE/ACM Int. Conf. Automated Softw. Eng. (ASE)*, Nov. 2019, pp. 974–985.

[P27] D. Manjah, S. Galland, C. De Vleeschouwer, and B. Macq, "Autonomous methods in multisensor architecture for smart surveillance," in *Proc. 16th Int. Conf. Agents Artif. Intell.*, 2024, pp. 823–831.

[P28] C. N. K. T. Melén, R. J. Machchhar, C. Wendin, and M. Bertoni, "An integrated simulation framework for system-of-systems value exploration," *Design Sci.*, vol. 11, p. e20, Jan. 2025.

[P29] A. Mendes, S. Loss, E. Cavalcante, F. Lopes, and T. Batista, "Mandala: An agent-based platform to support interoperability in systems-of-systems," in *Proc. IEEE/ACM 6th Int. Workshop Softw. Eng. Systems-of-Systems (SESoS)*, May 2018, pp. 21–28.

[P30] G. M. P. Wanderley, M.-H. Abel, E. C. Paraiso, and J. A. Barthès, "GAMBAD: A method for developing systems of systems," in *Proc. IEEE 30th Int. Conf. Tools With Artif. Intell. (ICTAI)*, Nov. 2018, pp. 813–817.

[P31] N. Naeem, P. Ratei, P. S. Prakasha, L. Asmer, R. Jaksche, H. Pak, K. Schweiger, A. Velieva, F. Naser, M. Swaid, J. Pertz, and M. Niklaß, "A collaborative system of systems simulation of urban air mobility," *CEAS Aeronaut. J.*, vol. 16, no. 3, pp. 729–744, Jul. 2025.

[P32] Y. Nomaguchi and K. Fujita, "A framework of system of systems design with scenario, multi-agent simulation and robustness assessment," *Proc. CIRP*, vol. 60, pp. 133–138, Jan. 2017.

[P33] F. Oquendo, "Architecting exogenous software-intensive systems-of-systems on the Internet-of-Vehicles with SosADL," *Syst. Eng.*, vol. 22, no. 6, pp. 502–518, Nov. 2019.

[P34] F. Oquendo, "Fuzzy mediating control systems for automating vehicle driving maneuvers: The overtaking case," in *Proc. TENCON IEEE Region 10 Conf. (TENCON)*, Oct. 2023, pp. 789–794.

[P35] P. Palensky, A. van der Meer, C. Lopez, A. Joseph, and K. Pan, "Applied cosimulation of intelligent power systems: Implementing hybrid simulators for complex power systems," *IEEE Ind. Electron. Mag.*, vol. 11, no. 2, pp. 6–21, Jun. 2017.

[P36] P. Patolla, D. Reichelt, D. Mothes, and G. Schneider, "An architecture for an automatic integration of IO-link sensors into a system of systems," in *Proc. IECON 47th Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2021, pp. 1–6.

[P37] Q. Qu, X. He, H. Cheng, and W. Zhang, "The design of SoS multi-agent system model based on emergence imitation," in *Proc. Int. Conf. Appl. Math., Modeling Simulation (AMMS)*, 2017, pp. 215–219.
[P38] R. Raman and A. Murugesan, "Framework for complex SoS emergent behavior evolution using deep reinforcement learning," in *Proc. INCOSE Int. Symp.*, 2022, vol. 32, no. 1, pp. 809–823.
[P39] R. Raman and A. Murugesan, "Reinforcement learning based system-of-systems approach for UAV swarms behavioral evolution," in *Proc. IEEE Int. Syst. Conf. (SysCon)*, Apr. 2022, pp. 1–8.
[P40] A. K. Raz, C. R. Kenley, and D. A. DeLaurentis, "A system-of-systems perspective for information fusion system design and evaluation," *Inf. Fusion*, vol. 35, pp. 148–165, May 2017.
[P41] M. A. Regueiro, J. R. R. Viqueira, C. Stasch, and J. A. Taboada, "Semantic mediation of observation datasets through sensor observation services," *Future Gener. Comput. Syst.*, vol. 67, pp. 47–56, Feb. 2017.
[P42] Z. Ruiwen, S. Bifeng, P. Yang, and Y. Qijia, "Improved method for subsystems performance trade-off in system-of-systems oriented design of UAV swarms," *J. Syst. Eng. Electron.*, vol. 30, no. 4, pp. 720–737, Aug. 2019.
[P43] A. R. Sadik, B. Bolder, and P. Subasic, "A self-adaptive system of systems architecture to enable its ad-hoc scalability: Unmanned vehicle fleet–mission control center case study," in *Proc. 7th Int. Conf. Intell. Syst., Metaheuristics Swarm Intell.*, Apr. 2023, pp. 111–118.
[P44] M. Seitz, F. Gehlhoff, L. A. Cruz Salazar, A. Fay, and B. Vogel-Heuser, "Automation platform independent multi-agent system for robust networks of production resources in Industry 4.0," *J. Intell. Manuf.*, vol. 32, no. 7, pp. 2023–2041, Oct. 2021.
[P45] A. Sheikh and E. K. P. Chong, "Multi-agent reinforcement learning framework for optimizing smart cities as system of systems," *Syst. Eng.*, vol. 29, no. 1, pp. 3–19, Jan. 2026.
[P46] S. Shrivastava, S. Barat, S. Kausley, V. Kulkarni, and B. Rai, "River digital twin for water quality prediction," in *Proc. Winter Simulation Conf. (WSC)*, Dec. 2024, pp. 608–619.
[P47] S. Singh and J. Heard, "A human-aware decision making system for human-robot teams," in *Proc. 17th Annu. Syst. Syst. Eng. Conf. (SOSE)*, Jun. 2022, pp. 268–273.
[P48] J.-B. Soyez, G. Morvan, R. Merzouki, and D. Dupont, "Multilevel agent-based modeling of system of systems," *IEEE Syst. J.*, vol. 11, no. 4, pp. 2084–2095, Dec. 2017.
[P49] A. Umair, A. Clausen, Y. Demazeau, and B. N. Jørgensen, "Agent based coordination protocol for system of cyber-physical systems," *Energy Informat.*, vol. 4, no. S2, p. 32, Sep. 2021.
[P50] F. Villas, L. K. Franzén, C. Jouannet, K. Amadori, and I. Staack, "Concept of operations in an agent-based simulation: A system-of-systems approach," in *Proc. 34th Congr. Int. Council Aeronaut. Sci. (ICAS)*, 2024, pp. 9–13.
[P51] G. M. P. Wanderley, M.-H. Abel, J.-P. Barthès, and E. C. Paraiso, "An advanced collaborative environment for software development," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2016, pp. 002917–002922.
[P52] G. M. P. Wanderley, M.-H. Abel, E. C. Paraiso, and J.-P.-A. Barthès, "MBA: A system of systems architecture model for supporting collaborative work," *Comput. Ind.*, vol. 100, pp. 31–42, Sep. 2018.
[P53] H. Ye, C. Chi, G. Qin, and Y. Kuang, "Multi-agent game strategies for kill chain optimization in networked aerial combat systems," *IET Control Theory Appl.*, vol. 19, no. 1, Jan. 2025.
[P54] H. R. Younes, A. S. E. Ahmed, and E. K. Elsayed, "Enhancement belief function with ontology to solve the conflict for the system of systems," *Int. J. Intell. Eng. Syst.*, vol. 12, no. 6, pp. 100–113, Dec. 2019.
[P55] H. Yuan, B. Ren, T. Chen, and X. Luo, "Decentralized coordination of intelligent system of systems under partial observability," *Adv. Eng. Informat.*, vol. 65, May 2025, Art. no. 103286.
[P56] L. Zhang, F. Ye, K. Xie, P. Gu, X. Wang, Y. Laili, C. Zhao, X. Zhang, M. Chen, T. Lin, and Z. Chen, "An integrated intelligent modeling and simulation language for model-based systems engineering," *J. Ind. Inf. Integr.*, vol. 28, Jul. 2022, Art. no. 100347.
[P57] M. Mogahed and M. Mansouri, "An ontology-based architecture for interoperable healthcare systems-of-systems: Structure, interaction patterns, and covenant-based governance," *Systems*, vol. 14, no. 4, p. 376, Mar. 2026.
[P58] M. Torkjazi and A. K. Raz, "A systems engineering methodology for system of autonomous systems: Architecture and integration," *Syst. Eng.*, vol. 29, no. 2, pp. 169–194, Mar. 2026.



THYAGO FERREIRA VIEIRA is currently pursuing the M.Sc. degree in systems and computing with the Military Institute of Engineering (IME), Rio de Janeiro, Brazil. His research interests include interoperability mechanisms for systems-of-systems, non-intrusive integration architectures, and intelligent autonomous agents.



RONALDO R. GOLDSCHMIDT received the B.Sc. degree in mathematics from Fluminense Federal University, in 1990, the M.Sc. degree in computer systems from the Military Institute of Engineering, in 1992, and the D.Sc. degree in electrical engineering from the Pontifical Catholic University of Rio de Janeiro, in 2004. He is currently an Associate Professor with the Military Institute of Engineering. His research interests include artificial intelligence, machine learning, and data science.



MARIA C. R. CAVALCANTI received the D.Sc. degree in computer engineering from the Federal University of Rio de Janeiro (UFRJ), Brazil, in 2003. She is currently a Full Professor with the Department of Computer Engineering, Instituto Militar de Engenharia (IME), Brazil. Previously, she was a System Analyst with UFRJ, from 1985 to 2004. After receiving her D.Sc. degree, since 2004, she has been with IME, as a Professor and a Researcher involved with the Postgraduate Program on Systems and Computing (PGSC) and the Postgraduate Program on Defense Engineering (PGED). Also, she has been participating in research projects on those topics, with funding from several Brazilian government agencies, including CNPq, CAPES, FINEP, and FAPERJ. Her current research interests include metadata and ontologies for data interlinking, conceptual modeling, semantic web (of data), data modeling for NoSQL DBMS, more recently, cybersecurity, and command and control systems (C2).



RICARDO CHOREN received the Ph.D. degree in computer science from the Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil. He is currently a Full Professor with the Military Institute of Engineering (IME), Brazil. His research interests include software and agentic engineering, with a focus on modeling and verification of complex environments.

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